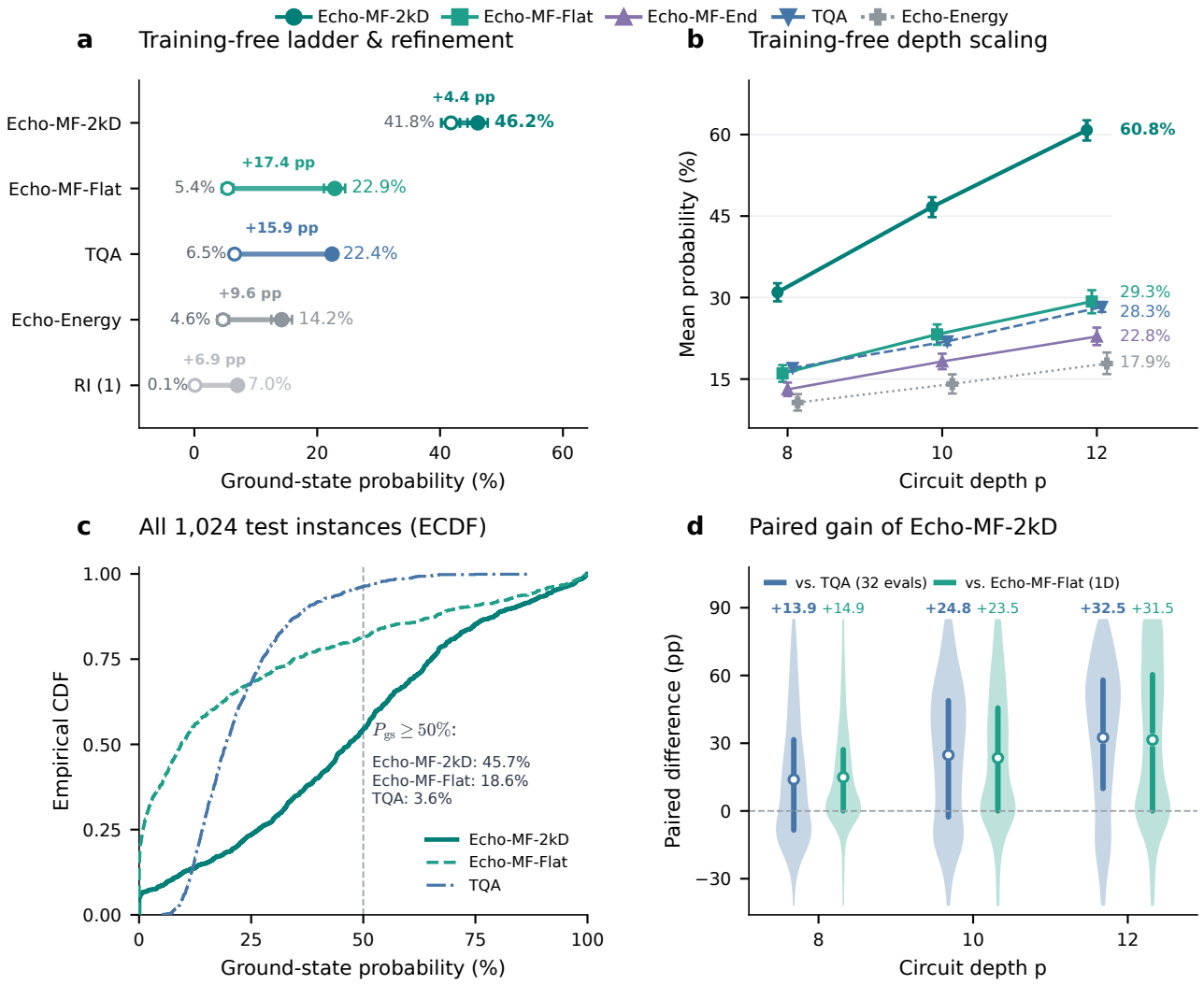

Echo-MF-2kD
Echo-MF-Flat
Echo-MF-End
TQA
Echo-Energy
a
Training-free ladder & refinement
+4.4 pp
41.8%
46.2%
+17.4 pp
5.4%
22.9%
+15.9 pp
6.5%
22.4%
+9.6 pp
4.6%
14.2%
+6.9 pp
0.1%
7.0%
Echo-MF-2kD
Echo-MF-Flat
TQA
Echo-Energy
RI (1)
0
20
40
60
Ground-state probability (%)
b
Training-free depth scaling
Mean probability (%)
60
45
30
15
60.8%
29.3%
28.3%
22.8%
17.9%
8
10
12
Circuit depth p
c
All 1,024 test instances (ECDF)
Empirical CDF
1.00
0.75
0.50
0.25
0.00
$P_{gs} \geq 50\%$:
Echo-MF-2kD: 45.7%
Echo-MF-Flat: 18.6%
TQA: 3.6%
Echo-MF-2kD
Echo-MF-Flat
TQA
0
25
50
75
100
Ground-state probability (%)
d
Paired gain of Echo-MF-2kD
vs. TQA (32 evals)
vs. Echo-MF-Flat (1D)
+13.9
+14.9
+24.8
+23.5
+32.5
+31.5
Paired difference (pp)
90
60
30
0
−30
8
10
12
Circuit depth p